12. SimTalk Reference

SimTalk Reference

Introduces *SimTalk*, the programming language that is integrated in *Plant Simulation*. It enables you to implement custom behavior and logic in your simulation model which is not covered by the built-in functionality of the simulation objects.

Remarks

SimTalk contains an *Interpreter*, which directly executes the instructions you wrote in objects of type **Method**, when *Plant Simulation* runs the simulation.

SimTalk is tightly integrated with the simulation objects. With *SimTalk* you can access all attributes, read-only attributes, and methods of the built-in objects.

Note

The **Copilot**, which accesses the *Plant Simulation Help* can assist you in writing your source code in a *Method*.

The *Copilot (local)* provides a better knowledge base and the ability to hook up an own LLM to solve your programming tasks.

You can also create your own reusable objects in an object of type **Dialog** with behavior for your specific needs and implemented in *SimTalk*. Together with the integrated **Method Debugger** *SimTalk* is an easy-to-use programming environment.

Instead of programming code in *SimTalk*, you can also program code in *Python*.

Note

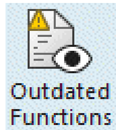
Python code is not intended to replace *SimTalk code*. As *SimTalk code* also accesses compiled built-in functions, it generally is quite a bit faster in execution than *Python code*.

As many university graduates are exposed to *Python*, writing *Python code* during their studies, the **PythonModule** makes programming in *Plant Simulation* easier and faster.

In *Plant Simulation 12.1* we introduced *SimTalk 2.0*. We simplified the syntax and added a number of features to make it easier to write your code.

The *Plant Simulation Help* only describes *SimTalk 2.0*. The topic **SimTalk 2.0 and SimTalk 1.0 Compared** describes the major differences.

SimTalk 1.0 stills works in the current and previous versions of *Plant Simulation* and you can continue using *SimTalk 1.0* in old and new models.

The command **Find Outdated Functions** 

on the **Debugger** ribbon tab finds outdated *methods*, *attributes*, *read-only attributes*, and *functions* in the source code of all *Methods* in your simulation model.

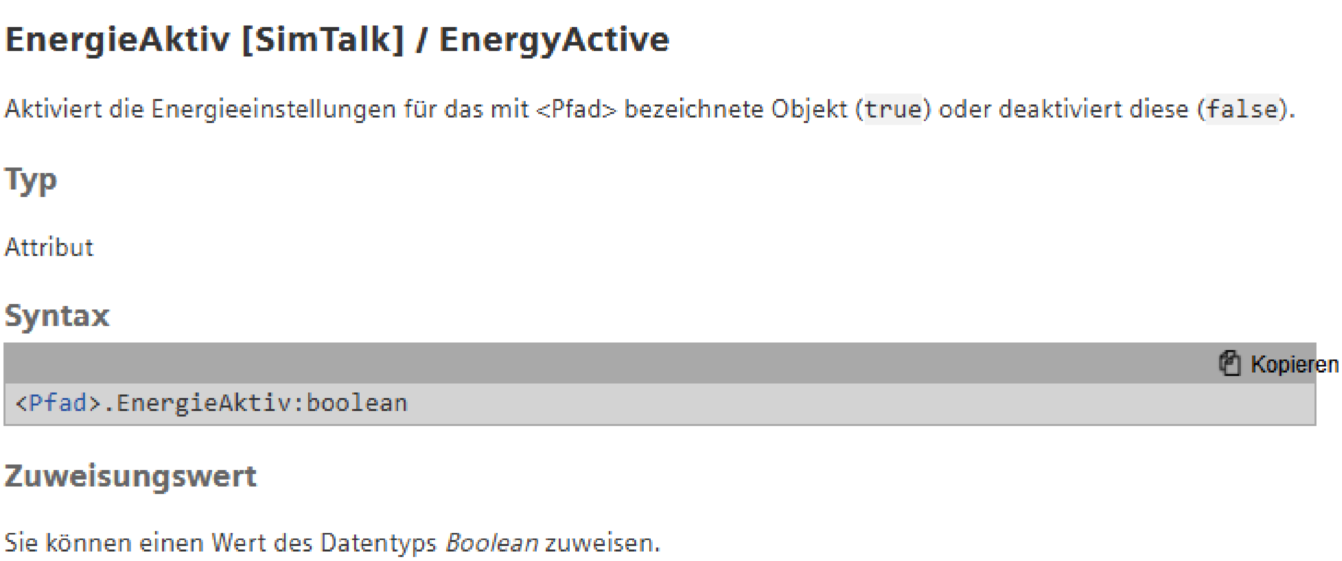
The description of *SimTalk* is divided in:

- **General Access to SimTalk**
- **SimTalk Access to 3D Functions**

Note

SimTalk supports **English** and **German** for programming the source code of your methods.

For this reason the **German Help** shows the **English** name of the *attribute*, *read-only attribute*, and *method* next to the **German** name, separated by a forward slash, for example **EnergieAktiv [SimTalk] / EnergyActive**. This facilitates programming methods if you are working with the **German** version of *Plant Simulation* and are creating an **English** language model that your colleagues in other countries can use as well.



EnergieAktiv [SimTalk] / EnergyActive

Aktiviert die Energieeinstellungen für das mit <Pfad> bezeichnete Objekt (`true`) oder deaktiviert diese (`false`).

Typ

Attribut

Syntax

Kopieren

```
<Pfad>.EnergieAktiv:boolean
```

Zuweisungswert

Sie können einen Wert des Datentyps *Boolean* zuweisen.

Compare **Creating a German Model with English Identifiers**.

See also

What's New

Introducing Plant Simulation

Update Arbitrary Old Models

The **User Interface Components**

The **Step-by-Step Help**

Outdated SimTalk Names

The **Quick Reference Card**

The **Add-Ins Reference Help**

The **Libraries Reference**